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Method for detecting methylation of SDC2 gene

Abstract

The present invention relates to a method of detecting methylation of an SDC2 gene, a composition for detecting methylation of an SDC2 gene, and a kit comprising same and, more particularly, to: a method of detecting methylation of an SDC2 gene by using primers for specifically amplifying a methylated SDC2 gene and a probe capable of complementarily hybridizing with a methylated SDC2 gene that has been specifically amplified by the primers; a composition for detecting methylation of an SDC2 gene; and a kit comprising same.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This is a U.S. national phase under 35 USC § 371 of International Patent Application No. PCT/KR2020/006692 filed May 22, 2020, which in turn claims priority under 35 USC § 119 of Korean Patent Application No. 10-2019-0072080 filed Jun. 18, 2019. The disclosures of all such applications are hereby incorporated herein by reference in their respective entireties, for all purposes.

REFERENCE TO SEQUENCE LISTING SUBMITTED VIA PATENT CENTER

(2) This application includes an electronically submitted sequence listing in .txt format. The .txt file contains a sequence listing entitled “602_UpdatedSeqListing_ST25.txt” created on May 12, 2025 and is 229,401 bytes in size. The sequence listing contained in this .txt file is part of the specification and is hereby incorporated by reference herein in its entirety.

TECHNICAL FIELD

(3) The present invention relates to a method of detecting methylation of an SDC2 gene, a composition for detecting methylation of an SDC2 gene, and a kit comprising the same, and more particularly to a method of detecting methylation of an SDC2 gene using a primer specifically amplifying a methylated SDC2 gene and a probe capable of complementary hybridization to the methylated SDC2 gene specifically amplified by the primer, a composition for detecting methylation of an SDC2 gene, and a kit comprising the same.

BACKGROUND ART

(4) The genomic DNA of mammalian cells, has a fifth base in addition to A, C, G, and T, which is 5-methylcytosine (5-mC), in which a methyl group is attached to the fifth carbon of a cytosine ring. 5-mC is always attached only to C of a CG dinucleotide (5'-mCG-3'), and this CG is often denoted as CpG. C in CpG is mostly methylated, with a methyl group attached thereto. This methylation of CpG inhibits the expression of repetitive sequences in the genome, such as Alu or transposons, and CpG is the site where extragenic changes most frequently occur in mammalian cells. 5-mC of this CpG is naturally converted into T through deamination. Accordingly, CpG in the mammalian genome appears only with a frequency of 1%, which is much lower than a normal frequency ($\frac{1}{4} \times \frac{1}{4} = 6.25\%$).

(5) There is a region in which CpGs are exceptionally dense, which is called a CpG site (CpG island). The CpG site is 0.2-3 kb in length, and is a highly concentrated region in which the distribution percentage of C and G bases is greater than 50% and the distribution percentage of CpG is 3.75% or more. About 45,000 CpG sites appear in the entire human genome, and are intensively found in the promoter region, which regulates gene expression. Indeed, CpG sites appear in promoters of housekeeping genes, which account for about half of human genes (Cross, S. et al., Curr. Opin. Gene Develop., 5:309, 1995). Abnormal DNA methylation is known to occur mainly in the 5' regulatory region of the corresponding gene, thereby reducing expression of the corresponding gene.

(6) On the other hand, in somatic cells of normal persons, the CpG islands of these housekeeping gene promoter regions are not methylated, but imprinted genes and inactivated genes on the X chromosome are methylated so as to prevent the expression thereof during development.

(7) During the carcinogenesis process, methylation occurs in the promoter CpG island, and expression of the corresponding gene is impaired. In particular, when methylation occurs in the promoter CpG islands of tumor suppressor genes, which regulate cell cycles or apoptosis, repair DNA, participate in cell adhesion and intercellular interaction, and inhibit invasion and metastasis, the expression and function of these genes are blocked, like mutations in coding sequences, thereby promoting the development and progression of cancer. Partial methylation may also appear on CpG islands due to aging.

(8) Promoter methylation of tumor-related genes is an important indicator of cancer, so it may be used in various ways, such as diagnosis and early diagnosis of cancer, prediction of cancer risk, prediction of cancer prognosis, follow-up after treatment, prediction of response to chemotherapy, and the like. Indeed, recent attempts have been actively made to investigate the promoter methylation of tumor-related genes in the blood, sputum, saliva, stool, urine, and the like, and to use the results thereof in the treatment of various types of cancer (Ahlquist, D. A. et al., Gastroenterol., 119:1219, 2000).

(9) Against this technical background, the inventors of the present application have ascertained that methylation of an SDC2 gene may be detected with high detection limit and accuracy using a primer specifically amplifying a methylated SDC2 gene and a probe capable of complementary hybridization to the methylated SDC2 gene specifically amplified by the primer, thus the present invention has been completed.

DISCLOSURE

(10) It is an object of the present invention to provide a method of detecting methylation of an SDC2 gene using a primer and a probe.

- (11) It is another object of the present invention to provide a composition for detecting methylation of an SDC2 gene including a primer and a probe.
- (12) It is still another object of the present invention to provide a kit for detecting methylation of an SDC2 gene including the composition.
- (13) In order to accomplish the above objects, the present invention provides a method of detecting methylation of an SDC2 gene comprising (a) treating a sample with at least one reagent differently modifying a methylated SDC2 gene and a non-methylated SDC2 gene, (b) performing treatment with a primer specifically amplifying the methylated SDC2 gene, and (c) performing treatment with a probe capable of complementary hybridization to the methylated SDC2 gene specifically amplified by the primer in step (b).
- (14) In addition, the present invention provides a composition for detecting methylation of an SDC2 gene comprising at least one reagent differently modifying a methylated SDC2 gene and a non-methylated SDC2 gene, a primer specifically amplifying the methylated SDC2 gene, and a probe capable of complementary hybridization to the methylated SDC2 gene specifically amplified by the primer.
- (15) In addition, the present invention provides a kit for detecting methylation of an SDC2 gene comprising the composition.
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Description

DESCRIPTION OF DRAWING

(1) FIG. 1 shows the results of verification of methylation of multiple primer sets on stool DNA.

MODE FOR INVENTION

(2) Unless otherwise defined, all technical and scientific terms used herein have the same meanings as those typically understood by those skilled in the art to which the present invention belongs. Generally, the nomenclature used herein is well known in the art and is typical.

(3) The inventors of the present application designed methylation-specific detection primers and probes capable of representing the entire CpG island of the SDC2 gene, and ascertained that methylation may be specifically detected only in methylated DNA through methylation-specific amplification. In addition, the ability of the SDC2 gene to diagnose colorectal cancer in colorectal cancer tissues, stool, and blood was evaluated using methylation-specific detection primers and probes. Based on the results thereof, it was confirmed that the sensitivity and specificity for the diagnosis of colorectal cancer were very high, so usefulness in the diagnosis of colorectal cancer was high.

(4) Accordingly, an aspect of the present invention pertains to a method of detecting methylation of an SDC2 gene comprising (a) treating a sample with at least one reagent differently modifying a methylated SDC2 gene and a non-methylated SDC2 gene, (b) performing treatment with a primer specifically amplifying the methylated SDC2 gene, and (c) performing treatment with a probe capable of complementary hybridization to the methylated SDC2 gene specifically amplified by the primer in step (b).

(5) According to the present invention, the step (a) is treating the sample containing target DNA with at least one reagent differently modifying the methylated DNA region and the non-methylated DNA region.

(6) As used herein, the term “methylation” refers modification into 5-methylcytosine (5-mC) in which a methyl group is attached to the fifth carbon of a cytosine base ring, and 5-methylcytosine is always attached only to C of the CG dinucleotide (5'-mCG-3'), and this CG is often referred to as CpG. Methylation of CpG inhibits the expression of repetitive sequences in the genome, such as Alu or transposons, and CpG is the site where extragenic changes most frequently occur in mammalian cells. 5-mC of this CpG is naturally converted into T through deamination, and thus,

CpG in the mammalian genome is present only at a frequency of 1%, which is much lower than a normal frequency ($\frac{1}{4} \times \frac{1}{4} = 6.25\%$).

(7) There is a region in which CpGs are exceptionally dense, which is called a CpG island. The CpG island is 0.2-3 kb in length, and is a highly concentrated site in which the distribution percentage of C and G bases is greater than 50% and the distribution percentage of CpG is 3.75% or more. About 45,000 CpG islands appear in the entire human genome, and are intensively found in the promoter region, which regulates gene expression. CpG islands actually appear in promoters of housekeeping genes, which account for about half of human genes.

(8) The nucleic acid isolated from a specimen is obtained from a biological sample of the specimen. In order to diagnose colorectal cancer or the stage of progression of colorectal cancer, the nucleic acid has to be isolated from colorectal tissue by scraping or biopsy. Such a sample may be obtained by various medical procedures known in the art.

(9) The extent of methylation of the nucleic acid of the sample obtained from the specimen is measured through comparison with the same portion of the nucleic acid from a specimen without a colorectal tissue cell growth abnormality. Hypermethylation indicates the presence of a methylated allele in at least one nucleic acid. When the same nucleic acid is tested in a specimen without a colorectal tissue cell growth abnormality, the methylation allele does not appear.

(10) “Normal” cells are cells that do not show abnormal cell morphology or a change in cytological properties. “Tumor” cells are cancer cells, and “non-tumor” cells are cells that are part of the diseased tissue but are not the site of the tumor.

(11) According to the present invention, early diagnosis of cell growth abnormalities in the colorectal tissue of a specimen is possible by determining the methylation stage of at least one nucleic acid isolated from the specimen. The methylation stage of at least one nucleic acid may be compared with the methylation stage of at least one nucleic acid isolated from a specimen not exhibiting abnormal colorectal tissue cell growth. Preferably, the nucleic acid is a CpG-containing nucleic acid such as a CpG island.

(12) According to the present invention, it is possible to diagnose a predisposition to cell growth abnormalities in the colorectal tissue of a specimen, including determining the methylation of at least one nucleic acid isolated from the specimen. The methylation stage of at least one nucleic acid may be compared with the methylation stage of at least one nucleic acid isolated from a specimen having no predisposition to abnormal cell growth in colorectal tissue.

(13) As used herein, the term “predisposition” refers to the property of being susceptible to the above-mentioned cell growth abnormality. A specimen having a predisposition is a specimen which does not yet exhibit a cell growth abnormality, but in which a cell growth abnormality is present or the likelihood of developing a cell growth abnormality is increased.

(14) The presence of CpG methylation in target DNA may be an indicator of a disease, and, for example, CpG methylation of any one of a promoter, a 5' untranslated region, and an intron of target DNA may be measured.

(15) The CpG-containing gene is typically DNA. However, the method of the present invention may be performed using a sample containing, for example, DNA, or DNA and RNA including mRNA, in which the DNA or RNA may be single-stranded or double-stranded, or a sample containing a DNA-RNA hybrid may be used.

(16) A nucleic acid mixture may also be used. As used herein, the term “multiple” includes both the case in which there is a plurality of specific nucleic acid sequence sites to be detected in a kind of gene and the case in which a plurality of target DNA sequences is included in one tube (a single reactor). The specific nucleic acid sequence to be detected may be a fraction of a large molecule, or may be present initially in the form of a discrete molecule in which the specific sequence constitutes the entire nucleic acid sequence. The nucleic acid sequence need not be a nucleic acid present in a pure form, and the nucleic acid may be a minor fraction of a complex mixture, such as one contained in whole human DNA.

(17) Particularly, the present invention is directed to detecting methylation of a plurality of target DNA sequences in a sample in a single reactor, in which the sample may include multiple target DNA sequences, and any target DNA may be used without limitation, so long as it is a gene that affects the development or progression of cancer when the expression thereof is suppressed due to abnormal methylation, as well as a control gene.

(18) In the present invention, the sample may be derived from a human body, and the sample may include, for example, colorectal cancer tissue, cells, stool, urine, blood, serum, or plasma.

(19) At least one reagent differently modifying the methylated DNA and the non-methylated DNA may be used without limitation, so long as it is able to distinguish between the non-methylated cytosine base and the methylated cytosine base, and examples of the reagent may include, but are not limited to, bisulfite, hydrogen sulfite, disulfite, and combinations thereof. Particularly, the cytosine base methylated by the reagent is not converted, and the non-methylated cytosine base may be converted into uracil or a base other than cytosine.

(20) In the present invention, step (b) is performing treatment with a primer specifically amplifying the methylated SDC2 gene.

(21) The primer may include at least one CpG dinucleotide. For example, for PCR, forward and reverse primers may be paired and used simultaneously. The forward primer may include, for example, a sequence selected from the group consisting of SEQ ID NOS: 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, and 34 to 1140. The reverse primer may include, for example, a sequence selected from the group consisting of SEQ ID NOS: 2, 5, 8, 11, 14, 17, 20, 23, 26, 29, 32, and 1141 to 1159. The particular primer pair for the primer that specifically amplifies the methylated SDC2 gene is set forth in Table 1 of Example 1 and in Table 5 of Example 4.

(22) In the present invention, step (c) is performing treatment with a probe capable of complementary hybridization to the methylated SDC2 gene specifically amplified by the primer.

(23) In a hybridization reaction, the conditions used to achieve a certain stringent level vary depending on the properties of the nucleic acid to be hybridized. For example, the length of the nucleic acid site to be hybridized, the extent of homology, the nucleotide sequence composition (e.g. GC/AT ratio), and the nucleic acid type (e.g. RNA, DNA) are taken into consideration in selecting the hybridization conditions. An additional consideration is whether the nucleic acid is immobilized on, for example, a filter or the like.

(24) Examples of very stringent conditions are as follows: 2×SSC/0.1% SDS at room temperature (hybridization conditions), 0.2×SSC/0.1% SDS at room temperature (low-stringency conditions), 0.2×SSC/0.1% SDS at 42° C. (moderate-stringency conditions), and 0.1×SSC at 68° C. (high-stringency conditions). The washing process may be performed using any one of these conditions, and, for example, high-stringency conditions or each of the above conditions may be used. The conditions may be applied for 10 to 15 minutes each time in the order described above, or all or some of the conditions described above may be repeatedly applied. As described above, however, the optimal conditions vary depending on the special hybridization reaction involved, and may be determined experimentally. Generally, high-stringency conditions are used for the hybridization of the probe of interest.

(25) The probe may include, for example, at least one CpG dinucleotide. Particularly, the probe may include a sequence selected from the group consisting of SEQ ID NOS: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, and 1160 to 1178.

(26) In some cases, the probe may be detectably labeled, and may be labeled with, for example, a radioactive isotope, a fluorescent compound, a bioluminescent compound, a chemiluminescent compound, a metal chelate, or an enzyme. Appropriate labeling of the probe as described above is a technique well known in the art, and may be performed through a typical method.

(27) The amount of the amplification product may be detected based on a fluorescence signal. The detection method may include an intercalating method using an intercalator that exhibits fluorescence by binding to the double-stranded DNA of the amplification product to which the

probe is bound, a method of using an oligonucleotide in which the 5' end is labeled with a fluorescent material and the 3' end is labeled with a quencher, or the like.

(28) The amplification according to the present invention may be performed through real-time quantitative amplification, for example, real-time polymerase chain reaction (PCR), and in real-time PCR, the amount of a PCR amplification product may be detected using a fluorescence signal. As real-time PCR proceeds, the intensity of the fluorescence signal increases in proportion to an increase in the amount of polynucleotide, and an amplification profile curve showing the intensity of the fluorescence signal depending on the number of amplification cycles is obtained.

(29) In general, the amplification profile curve is divided into a baseline region which shows a fluorescence signal in the background that does not substantially reflect the amount of polynucleotide, an exponential region in which the fluorescence signal increases with an increase in the amount of a polynucleotide product, and a plateau region in which PCR reaches saturation and thus the intensity of the fluorescence signal no longer increases.

(30) Typically, the fluorescence signal intensity at the transition point from the baseline region to the exponential region, namely at the point when the amount of a PCR amplification product reaches an amount detectable by fluorescence, is referred to as a threshold, and the number of amplification cycles corresponding to the threshold value on the amplification profile curve is referred to as a threshold cycle (Ct) value.

(31) By measuring the Ct value, analyzing the standard curve in which the concentration is determined based on the Ct (threshold cycle) value for a standard material, and confirming the concentration of the amplified gene, the methylation-specific sensitivity and/or specificity may be determined.

(32) In one embodiment, the methylation may be detected using any method selected from the group consisting of PCR, methylation-specific PCR, real-time methylation-specific PCR, PCR using a methylated-DNA-specific binding protein, PCR using a methylated-DNA-specific binding antibody, quantitative PCR, gene chip, sequencing, sequencing by synthesis, and sequencing by ligation.

(33) (1) Methylation-specific PCR: For detection by methylation-specific PCR, when treated with a bisulfate, the cytosine in the 5'-CpG'-3' region remains as cytosine in the case of methylation, and is converted into uracil in the case of non-methylation. Therefore, a primer corresponding to a region in which the 5'-CpG-3' nucleotide sequence exists may be prepared for the nucleotide sequence converted after treatment with bisulfite. When PCR is performed using primers, in the case of methylation, a PCR product is made due to the use of the primers corresponding to the methylated nucleotide sequence, and methylation may be confirmed through agarose gel electrophoresis. Here, the methylation detection probe may be TaqMan, Molecular Beacon, or a probe having a self-reporting function or an energy-transfer labeling function, but is not limited thereto.

(34) (2) Real-time methylation-specific PCR: Real-time methylation-specific PCR is a real-time measurement method modified from methylation-specific PCR, and includes treating genomic DNA with bisulfite, designing PCR primers corresponding to the methylated nucleotide sequence, and performing real-time PCR using the primers. Here, there are two detection methods: a detection method using a TaqMan probe complementary to the amplified nucleotide sequence and a detection method using SYBR Green. Therefore, real-time methylation-specific PCR is capable of selectively quantitatively analyzing only methylated DNA. As such, a standard curve is created using an in-vitro methylated DNA sample, and a gene having no 5'-CpG-3' sequence in the nucleotide sequence is also amplified as a negative control for standardization, thus quantitatively analyzing the extent of methylation.

(35) (3) PCR using methylated-DNA-specific binding protein, quantitative PCR, and DNA chip assay: In the PCR using a methylated-DNA-specific binding protein or the DNA chip method, when a protein that specifically binds only to methylated DNA is mixed with DNA, the protein

specifically binds only to methylated DNA, so methylated DNA may be selectively isolated.

(36) In addition, methylation may be measured through quantitative PCR, and methylated DNA isolated with the methylated-DNA-specific binding protein is labeled with a fluorescent dye and hybridized to a DNA chip integrated with complementary probes, thereby measuring methylation.

(37) (4) Detection of differential methylation-bisulfite sequencing method: Another method of detecting a nucleic acid containing methylated CpG includes bringing a nucleic-acid-containing sample into contact with an agent that modifies non-methylated cytosine and amplifying the CpG-containing nucleic acid in the sample using CpG-specific oligonucleotide primers. Here, the oligonucleotide primers may be characterized in that the methylated nucleic acid is detected by distinguishing between modified methylated and non-methylated nucleic acids. The amplification step is optional and preferable, but not essential. The method relies on the PCR reaction to distinguish between modified (e.g. chemically modified) methylated DNA and non-methylated DNA.

(38) (5) Bisulfite sequencing method: Another method of detecting nucleic acid containing methylated CpG includes bringing a nucleic-acid-containing sample into contact with an agent that modifies non-methylated cytosine and amplifying the CpG-containing nucleic acid in the sample using methylation-independent oligonucleotide primers. Here, the oligonucleotide primers may be characterized in that the nucleic acid is amplified without distinguishing between modified methylated and non-methylated nucleic acids. The amplified product has been described in connection with bisulfite sequencing for detection of methylated nucleic acids by next-generation sequencing methods or for sequencing by the Sanger method using a sequencing primer.

(39) (6) Next-generation sequencing methods include a sequencing-by-synthesis method and a sequencing-by-ligation method. These methods are characterized in that, instead of creating a bacterial clone, a single DNA fragment is spatially separated, amplified in situ (clonal amplification), and sequenced. Here, since hundreds of thousands of fragments are read simultaneously, such a method is also called a massively parallel sequencing method.

(40) Basically, a sequencing-by-synthesis method is performed, a method of obtaining signals by sequentially attaching mono- or di-nucleotides is used, and examples thereof may include pyrosequencing, ion torrent, and Solexa methods.

(41) Examples of NGS devices based on the sequencing-by-synthesis method include Roche's 454 platform, Illumina's HiSeq platform, Life Technology's Ion PGM platform, and Pacific BioSciences' PacBio platform. 454 and Ion PGM use emulsion PCR as a clonal amplification method, and HiSeq uses bridge amplification. The sequencing-by-synthesis method reads the sequence by detecting phosphate, protons, or pre-attached fluorescence generated when DNA is synthesized by sequentially attaching one nucleotide. In the method of detecting the sequence, 454 uses a pyrosequencing method using phosphoric acid, and Ion PGM uses proton detection. HiSeq and PacBio detect fluorescence to decode the sequence.

(42) A sequencing-by-ligation method is a sequencing technique using DNA ligase, which identifies nucleotides at certain positions in a DNA nucleotide sequence. Unlike most sequencing techniques using a polymerase, the sequencing-by-ligation method does not use a polymerase and is characterized in that DNA ligase does not ligate mismatched sequences. An example thereof is the SOLiD system. In this technique, two bases are read with spacing, which is repeated five times independently through primer reset, so accuracy is improved by reading each base twice in duplicate.

(43) In the sequencing-by-ligation method, among the dinucleotide primer sets made of 16 combinations, dinucleotide primers corresponding to the nucleotide sequences are sequentially ligated, the combination of these ligations is finally analyzed, and the nucleotide sequence of the corresponding DNA is completed.

(44) Here, the next-generation sequencing method may be exemplified by a sequencing-by-synthesis method or a sequencing-by-ligation method. The methylated-DNA-specific binding

protein is not limited to MBD2bt, and the antibody is a 5'-methyl-cytosine antibody, but is not limited thereto.

(45) With regard to the primer used in the present invention, when a reagent such as bisulfite is used in step (a), the cytosine in the 5'-CpG'-3' site remains as cytosine in the case of methylation, and is converted into uracil in the case of non-methylation. Therefore, a primer corresponding to a region in which the 5'-CpG-3' nucleotide sequence exists may be prepared for the nucleotide sequence converted after treatment with a reagent, such as bisulfite.

(46) The primer may be designed to have "substantial" complementarity with each strand of the locus to be amplified in the SDC2 gene. This means that the primer has sufficient complementarity to hybridize with the corresponding nucleic acid strand under the conditions for the polymerization reaction.

(47) Another aspect of the present invention pertains to a composition for detecting methylation of an SDC2 gene including at least one reagent differently modifying a methylated SDC2 gene and a non-methylated SDC2 gene, a primer specifically amplifying the methylated SDC2 gene, and a probe capable of complementary hybridization to the methylated SDC2 gene specifically amplified by the primer.

(48) Since the components contained in the composition according to the present invention overlap the components described above, a description thereof is equally applied.

(49) Still another aspect of the present invention pertains to a kit for detecting methylation of target DNA including the composition described above.

(50) In one embodiment, the kit includes compartmentalized carrier means that accommodates a sample therein, a container including a reagent, a container including a primer capable of amplifying the SDC2 gene 5'-CpG-3', and a container including a probe for detecting the amplification product.

(51) The carrier means is suitable for accommodating one or more individual containers, such as bottles and tubes, containing independent components for use in the method of the present invention. In the specification of the present invention, one of ordinary skill in the art may readily determine the apportionment of the necessary agents in the containers.

(52) A better understanding of the present invention may be obtained through the following examples. These examples are merely set forth to illustrate the present invention, and are not to be construed as limiting the scope of the present invention, as will be apparent to those of ordinary skill in the art.

Example 1: Evaluation of Ability of SDC2 Gene to Diagnose Colorectal Cancer in Colorectal Cancer Tissue

(53) In order to evaluate the ability of the SDC2 gene to diagnose colorectal cancer, 11 sets of methylation-specific detection primers and probes capable of representing the entire CpG island of the SDC2 gene were designed (Table 1), and methylation-specific real-time PCR (qMSP) was performed. To this end, genomic DNA was isolated from the surgical tissue of 20 colorectal cancer patients using cancer tissue and normal tissue adjacent thereto (QIAAMP® DNA mini kit, Qiagen), and genomic DNA (2.0 µg) was treated with bisulfate using an EZ DNA methylation-Gold kit (Zymo Research, USA), dissolved in 10 µl of sterile distilled water, and used for methylation-specific real-time PCR (qMSP). qMSP was performed using bisulfite-treated genomic DNA as a template and using the methylation-specific primers and probes designed in Table 1 below. For qMSP, a ROTOR-GENE® Q PCR machine (Qiagen) was used. A total of 20 µl of a PCR reaction solution (20 ng of template DNA, 4 µl of 5×AptaTaq DNA Master (Roche Diagnostics), 2 µl (2 pmol/µl) of PCR primer, 2 µl (2 pmol/µl) of TaqMan probe, and 10 µl of D.W.) was prepared, and PCR was performed under conditions of 95° C. for 5 minutes followed by 95° C. for 15 seconds and an appropriate annealing temperature (58° C. to 61° C.) for 1 minute for a total of 40 cycles. Whether the PCR product was amplified was confirmed by measuring the cycle threshold (C.sub.T) value. Methylated and non-methylated control DNAs were tested along with the sample

DNA using an EPITECT® PCR control DNA set (Qiagen, cat. no. 59695). As an internal control gene, a COL2A1 gene (Kristensen et al., 2008) was used. The extent of methylation of each sample was measured using a C.sub.T (cycle threshold) value.

(54) The sensitivity and specificity for colorectal cancer diagnosis of each primer and probe set were calculated through ROC curve analysis (MedCalc program, Belgium) using the C.sub.T values of colorectal cancer tissue and normal tissue adjacent thereto (Table 2).

(55) TABLE-US-00001 TABLE 1 Primer and probe sequences for SDC2 gene

qMSP Amplification product	size	SEQ Set	Primer Sequence (5'-->3') (bp)	ID	NO:
SDC2 F	111	1	i-1 R CTCGATACCCCCATTCCCGCG	2	Probe
AAGCGTTGTTTCGTTGGCGTTATTTTCGCGG	3	SDC2 F	GCGTTATTTTCGCGGTTTCGC	114	4
i-2 R CCACGCAACAAAACCCGCCG	5	Probe	CGAGAATTGCGGTTTGGTTTAGT	6	
CGTAGAG SDC2 F	108	7	i-5 R		
CGACCCGAAACCGAACGCCG	8	Probe	CGGGAGTTCGTAAGTAGGGCGAGGCG	9	SDC2 F
GGCGGGGTACGTGTGATAC	98	10	i-6 R GCCAAAACCAAAAAAACGAACGTAACG	11	
Probe CGGTTTCGGGTCGTTTGGTCGTTGG	12	SDC2 F			
GATAGAGGTTTTTTTTTCGTTACGTTTC	96	13	i-7 R CGACAAACGCTCCGCCGAAAACG	14	
Probe TTTGGCGGGGCGTTTTTGGGGTCGGGA	15	SDC2 F	CGGGTTCGCGAGGGAAC	105	
16	i-8 R	CTTACATACAATCAAACAAAAAAA	17	CTAACGCG	Probe
CGCGCGTTTTTCGAGATTAGGGAT	18	GATTTG	SDC2 F	CGTTTTGGCGGTGGGAATTTG	
93	19	i-9 R	ACGCCCAAATAAAAAACAACACTACGAACG	20	Probe
TGGTCGCGTTTTTCGGGGGTTGGAG	21	SDC2 F	GATTCGGGGAGGGAGGC	128	22
t-10 R		ACGACGAAAACGCGCGATCCG	23	Probe	CGTAGTCGCGGAGTTAGTGGTTTCGTT
24		SDC2 F	GTCGGTGAGTAGAGTCGGC	122	25
p-11 R		CGACAATATAACTCCCAAATAAACCCG	26	Probe	CGGAGTCGCGGCGTTTATTGGTTTTTC
27	SDC2 F	CGGAGGATGCGCGCGTC	129	bp	28
i-12 R		CGAACGAACGCATTTATCCCAACG	29	Probe	AGTTACGAGAGGAGTTCGTAGGGAA
30		TAGG	SDC2 F	CGGTTAGGGCGAGGTAATCG	122
bp	31	i-13 R			
CGACGTCCCAACATTTTCGAACG	32	Probe	CGTGGAATCGTAGTAGGCGATTTTTT	33	
TAAGG					

(56) Based on the results of verification of SDC2 gene methylation using colorectal cancer tissue and normal tissue DNA adjacent thereto, the sensitivity for colorectal cancer diagnosis was 80% (16/20) to 95.0% (19/20) and the specificity therefor was 85.0% (3/20) to 95.0% (1/20), which was evaluated to be superior. Therefore, it was confirmed that the usefulness of SDC2 gene methylation in the diagnosis of colorectal cancer was high.

(57) TABLE-US-00002 TABLE 2 Evaluation of ability of SDC2 gene to diagnose colorectal cancer in colorectal cancer tissue

Primer and Sensitivity (%)	Specificity(%)	probe set	Cut-off (C.sub.T)	n	
= 20	n = 20	SDC2i-1	<35	80 95	
SDC2i-2	95 95	SDC2i-5	85 90	SDC2i-6	90 95
SDC2i-7	90 85	SDC2i-8	80 90	SDC2i-9	80 90
SDC2t-10	80 95	SDC2p-11	85 90	SDC2i-12	90 95
SDC2i-13	95 90				

Example 2: Evaluation of Ability of SDC2 Gene to Diagnose Colorectal Cancer Using Stool DNA

(58) In order to evaluate the ability of the SDC2 gene to diagnose colorectal cancer, methylation-specific real-time PCR (qMSP) was performed using the SDC2 gene methylation-specific detection primers and probes (Table 1) described in Example 1. To this end, genomic DNA was isolated from stool DNA of 20 colorectal cancer patients (Yonsei Medical Center Severance Hospital) and 20 normal persons (Yonsei Medical Center Severance Hospital check-up) (Stool DNA mini kit, Qiagen), and genomic DNA (2.0 µg) was treated with bisulfite using the EZ DNA methylation-Gold kit (Zymo Research, USA), dissolved in 10 µl of sterile distilled water, and used for methylation-specific real-time PCR (qMSP). qMSP was performed by the method described in Example 1.

(59) The sensitivity and specificity for colorectal cancer diagnosis of each primer and probe set were calculated through ROC curve analysis (MedCalc program, Belgium) using C.sub.T values

from stool DNA of colorectal cancer patients and normal persons (Table 3).

(60) TABLE-US-00003 TABLE 3 Evaluation of ability of SDC2 gene to diagnose colorectal cancer in stool DNA Primer and Sensitivity(%), Specificity(%), probe set Cut-off (C.sub.T) n = 20 n = 20 SDC2i-1 <40 80 90 SDC2i-2 90 90 SDC2i-5 80 85 SDC2i-6 90 90 SDC2i-7 80 85 SDC2i-8 80 90 SDC2i-9 80 85 SDC2t-10 80 90 SDC2p-11 85 90 SDC2i-12 80 90 SDC2i-13 85 90

(61) Based on the results of verification of SDC2 gene methylation using stool DNA from colorectal cancer patients (20 people) and normal persons (20 people), the sensitivity for colorectal cancer diagnosis was 80% (16/20) to 90.0% (18/20), and the specificity therefor was 85.0% (3/20) to 90.0% (2/20), which was evaluated to be superior. Therefore, it was confirmed that the usefulness of SDC2 gene methylation in the diagnosis of colorectal cancer in stool DNA was high.

Example 3: Evaluation of Ability of SDC2 Gene to Diagnose Colorectal Cancer in Blood

(62) In order to evaluate the ability of the SDC2 gene to diagnose colorectal cancer, methylation-specific real-time PCR (qMSP) was performed using the SDC2 gene methylation-specific detection primers and probes (Table 1) described in Example 1. To this end, DNA was isolated from 1 mL of serum of each of 10 colorectal cancer patients (Chungnam National University Hospital) and 10 normal persons (Innovative Research, USA) (Dynabead, Thermo Fisher), and the DNA was treated with bisulfate using the EZ DNA methylation-Gold kit (Zymo Research, USA), dissolved in 10 μ l of sterile distilled water, and used for methylation-specific real-time PCR (qMSP). qMSP was performed by the method described in Example 1.

(63) The sensitivity and specificity for colorectal cancer diagnosis of each primer and probe set were calculated through ROC curve analysis (MedCalc program, Belgium) using the C.sub.T values resulting from qMMSP using serum DNA from colorectal cancer patients and normal persons (Table 4).

(64) TABLE-US-00004 TABLE 4 Evaluation of ability of SDC2 gene to diagnose colorectal cancer in blood (serum) DNA Primer and Sensitivity(%), Specificity(%), probe set Cut-off (C.sub.T) n = 20 n = 20 SDC2i-1 <40 75 90 SDC2i-2 85 90 SDC2i-5 85 90 SDC2i-6 70 90 SDC2i-7 70 85 SDC2i-8 80 90 SDC2i-9 70 85 SDC2t-10 80 90 SDC2p-11 75 90 SDC2i-12 80 90 SDC2i-13 85 85

(65) Based on the results of verification of SDC2 gene methylation using blood DNA from colorectal cancer patients (10 people) and normal persons (10 people), the sensitivity for colorectal cancer diagnosis was 70% (14/20) to 90.0% (18/20), and the specificity therefor was 85.0% (3/20) to 90.0% (2/20), which was evaluated to be superior. Therefore, it was confirmed that the usefulness of SDC2 gene methylation in the diagnosis of colorectal cancer in blood DNA was high.

(66) Example 4: Evaluation of ability of SDC2 gene to diagnose colorectal cancer using stool DNA

(67) In order to evaluate the ability of the SDC2 gene to diagnose colorectal cancer, methylation-specific real-time PCR (qMSP) was performed using the SDC2 gene methylation-specific detection primers and probes (Table 1) described in Example 1. To this end, genomic DNA was isolated from stool DNA of 20 colorectal cancer patients (Yonsei Medical Center Severance Hospital) and 20 normal persons (Yonsei Medical Center Severance Hospital check-up) (Stool DNA mini kit, Qiagen), and the genomic DNA (2.0 μ g) was treated with bisulfite using the EZ DNA methylation-Gold kit (Zymo Research, USA), dissolved in 10 μ l of sterile distilled water, and used for methylation-specific real-time PCR (qMSP). qMSP was performed by the method described in Example 1.

(68) The positivity frequency for colorectal cancer diagnosis of each primer and probe set was calculated using C.sub.T values from stool DNA of colorectal cancer patients and normal persons (FIG. 1).

(69) As shown in FIG. 1, all primer sets exhibited negative methylation in normal persons upon colonoscopy (specificity: 100%). In colorectal cancer patients, the positive methylation frequency of the newly designed primers was higher than that of SEQ ID NO: 1179

(TAGAAATTAATAAGTGAGAGGGCGT) and SEQ ID NO: 1180

(GACTCAAACCTCGAAACTCGAA) SDC2 primer set described in Korean Patent No. 1142131

(in FIG. 1, the SDC2 item showed positive methylation frequency on the primers of Korean Patent No. 1142131).

(70) The positive methylation frequency in colorectal cancer patients of primer and probe sets is shown in Table 5 below.

(71) TABLE-US-00005 TABLE 5 Positive methylation frequency of primer and probe sets in stool DNA Positive methylation Primer and Cut-off frequency probe set (CT) (%), n = 20 SDC2* <40 45 SDC2i-1 55 SDC2i-2 65 SDC2i-5 65 SDC2i-6 55 SDC2i-7 65 SDC2i-8 60 SDC2i-9 65 SDC2t-10 60 SDC2p-11 55 SDC2i-12 60 SDC2i-13 80 *Positive methylation frequency in SEQ ID NO: 1127 (TAGAAATTAATAAGTGAGAGGGCGT) and SEQ ID NO: 1180 (GACTCAAACCTCGAAACTCGAA) SDC2 primer set described in Korean Patent No. 1142131

(72) Based on the results thereof, it was confirmed that all newly designed primer and probe sets exhibited high positive methylation frequency in colorectal cancer patients.

Example 5: Evaluation of Detection of SDC2 Gene Methylation Through Multiple Methylation-Specific Primer And Probe Design

(73) In order to evaluate the ability of the SDC2 gene to diagnose colorectal cancer, 1,107 sets of methylation-specific detection primers and probes capable of representing the entire CpG island of the SDC2 gene were designed (Table 6), and methylation-specific real-time PCR (qMSP) was performed. To this end, the abilities of these primers and probes to detect SDC2 gene methylation were evaluated using bisulfite-treated human methylated DNA and non-methylated DNA (EPITECT® PCR control DNA set, Qiagen, Cat. no. 59695). 20 ng of the DNA was dissolved in 10 µl of sterile distilled water and then used for methylation-specific real-time PCR (qMSP). For qMSP, a ROTOR-GENE® Q PCR machine (Qiagen) was used. A total of 20 µl of a PCR reaction solution (20 ng of template DNA, 4 µl of 5×AptaTaq DNA Master (Roche Diagnostics), 2 µl (2 pmol/µl) of PCR primer, 2 µl (2 pmol/µl) of TaqMan probe, and 10 µl of D.W.) was prepared, and PCR was performed under conditions of 95° C. for 5 minutes followed by 95° C. for 15 seconds and an appropriate annealing temperature (58° C. to 61° C.) for 1 minute for a total of 40 cycles. Whether the PCR product was amplified was confirmed by measuring the cycle threshold (CT) value. As an internal control gene, a COL2A1 gene (Kristensen et al., 2008) was used. For the extent of methylation of each sample, the sensitivity and specificity for colorectal cancer diagnosis of each primer and probe set were calculated through ROC curve analysis (MedCalc program, Belgium) using CT (cycle t) values.

(74) TABLE-US-00006 TABLE 6 Primer and probe sequences for SDC2 gene qMSP Set Primer Sequence (5'-->3') 1 to F1-F61 SEQ ID NOS: 34 to 94, 61 respectively R1 GAACGCATTTATCC (SEQ ID NO: 1141) P1 TTTAAGTATATATCGGAGATTCGTTG (SEQ ID NO: 1160) 62 to F62- SEQ ID NOS: 95 to 155, 122 F122 respectively R2 AAATAACGCCAACG (SEQ ID NO: 1142) P2 TTTTTTTTTTTTAGAAAAGCGTTGTT (SEQ ID NO: 1161) 123 to F123- SEQ ID NOS: 156 to 216, 183 F183 respectively R3 CAAAAACCTCTACG (SEQ ID NO: 1143) P3 TATCGAGAATTGCGGTTTGGTTTAGT (SEQ ID NO: 1162) 184 to F184- SEQ ID NOS: 217 to 277, 244 244 respectively R4 CTCCGTCCTTCCCA (SEQ ID NO: 1144) P4 TTTTCGGCGGGTTTTGTTCGTGGTT (SEQ ID NO: 1163) 245 to F245- SEQ ID NOS: 278 to 338, 305 F305 respectively R5 CGAAATAAAACCGT (SEQ ID NO: 1145) P5 GGAGTTTGGGTCGGGTTTCGCGAGGGA (SEQ ID NO: 1164) 306 to F306- SEQ ID NOS: 339 to 399, 366 F366 respectively R6 AATAATATACGAAA (SEQ ID NO: 1146) P6 TGATTTGGAAATTCGGGGTTTTTTT (SEQ ID NO: 1165) 367 to F367 to SEQ ID NOS: 400 to 480, 447 F447 respectively R7 AAACACTCGCGAAT (SEQ ID NO: 1147) P7 GGGGAGATGGGGGTTAGATTAAAGAG (SEQ ID NO: 1166) 448 to F448 to SEQ ID NOS: 481 to 541, 508 F508 respectively R8 ATTACCTCGCCCTA (SEQ ID NO:

25.1 N.D 135 24.9 N.D 136 28.4 N.D 137 24.3 N.D 138 24.3 N.D 139 25.1 N.D 140 24.9 N.D 141
25.9 N.D 142 27.6 N.D 143 24.3 N.D 144 24.3 N.D 145 28.3 N.D 146 25.3 N.D 147 26.4 N.D 148
27.4 N.D 149 26.3 N.D 150 25.2 N.D 151 25.7 N.D 152 27.6 N.D 153 27.8 N.D 154 29.3 N.D 155
25.4 N.D 156 27.4 N.D 157 26.3 N.D 158 25.2 N.D 159 25.7 N.D 160 27.6 N.D 161 27.8 N.D 162
29.3 N.D 163 25.4 N.D 164 25.7 N.D 165 27.4 N.D 166 28.2 N.D 167 27.2 N.D 168 24.2 N.D 169
27.9 N.D 170 28.6 N.D 171 28.4 N.D 172 24.4 N.D 173 24.2 N.D 174 25.1 N.D 175 24.9 N.D 176
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27.6 N.D 380 24.3 N.D 381 27.4 N.D 382 26.3 N.D 383 25.2 N.D 384 25.7 N.D 385 27.6 N.D 386
27.8 N.D 387 29.3 N.D 388 25.1 N.D 389 26.3 N.D 390 27.4 N.D 391 26.3 N.D 392 25.2 N.D 393
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25.2 N.D 625 25.7 N.D 626 27.6 N.D 627 27.8 N.D 628 29.3 N.D 629 25.4 N.D 630 25.7 N.D 631
27.4 N.D 632 28.2 N.D 633 27.2 N.D 634 24.2 N.D 635 27.4 N.D 636 26.3 N.D 637 25.2 N.D 638
25.7 N.D 639 27.6 N.D 640 27.8 N.D 641 29.3 N.D 642 24.2 N.D 643 27.9 N.D 644 28.6 N.D 645
28.4 N.D 646 24.4 N.D 647 24.2 N.D 648 25.1 N.D 649 24.9 N.D 650 25.9 N.D 651 27.9 N.D 652
26.1 N.D 653 24.8 N.D 654 25.5 N.D 655 25.7 N.D 656 24.9 N.D 657 24.2 N.D 658 25.5 N.D 659
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24.3 N.D 674 23.9 N.D 675 25.3 N.D 676 26.4 N.D 677 27.4 N.D 678 26.3 N.D 679 25.2 N.D 680
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27.2 N.D 1009 24.2 N.D 1010 27.9 N.D 1011 28.6 N.D 1012 28.4 N.D 1013 24.4 N.D 1014 24.2
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1078 27.6 N.D 1079 24.3 N.D 1080 24.3 N.D 1081 28.3 N.D 1082 25.3 N.D 1083 27.4 N.D 1084
26.3 N.D 1085 25.2 N.D 1086 25.7 N.D 1087 27.6 N.D 1088 27.8 N.D 1089 29.3 N.D 1090 27.9
N.D 1091 28.6 N.D 1092 28.4 N.D 1093 24.4 N.D 1094 24.2 N.D 1095 25.1 N.D 1096 24.9 N.D
1097 25.9 N.D 1098 27.6 N.D 1099 24.3 N.D 1100 27.4 N.D 1101 26.3 N.D 1102 25.2 N.D 1103
25.7 N.D 1104 27.6 N.D 1105 27.8 N.D 1106 29.3 N.D 1107 25.4 N.D

INDUSTRIAL APPLICABILITY

(77) The present invention has the effect of providing a method of conferring information for the diagnosis of colorectal cancer by detecting methylation of the CpG island of an SDC2 gene, which is a colorectal-cancer-specific marker gene, with high detection sensitivity. Since colorectal cancer can be diagnosed at the initial transformation stage, early diagnosis is possible, and the method of the present invention is capable of diagnosing colorectal cancer more accurately and quickly than typical methods and is thus useful.

(78) Although specific embodiments of the present invention have been disclosed in detail as described above, it will be obvious to those of ordinary skill in the art that the description is merely

of preferable exemplary embodiments, and is not to be construed as limiting the scope of the present invention. Therefore, the substantial scope of the present invention will be defined by the appended claims and equivalents thereof.

SEQUENCE LIST FREE TEXT

(79) An electronic file is attached.

Claims

1. A method of detecting methylation of an SDC2 gene, comprising: (a) treating genomic DNA from a stool sample with a reagent comprising bisulfite, hydrogen sulfite, modifies or disulfite, which differently modifies a methylated SDC2 gene and a non-methylated SDC2 gene; (b) performing treatment with a primer pair comprising the sequence of SEQ ID NOs: 31 and 32 specifically amplifying the methylated SDC2 gene; and (c) detecting the methylation of the SDC2 gene using a probe comprising the sequence of SEQ ID NO: 33 that hybridizes to the amplified methylated SDC2 gene specifically amplified in step (b).
 2. The method according to claim 1, wherein at least one cytosine base is converted into uracil through treatment with the reagent.
 3. The method according to claim 1, wherein the detecting the methylation is performed using a process selected from the group consisting of PCR, methylation-specific PCR, real-time methylation-specific PCR, PCR using a methylated-DNA-specific binding protein, PCR using a methylated-DNA-specific binding antibody, quantitative PCR, gene chip, sequencing, sequencing by synthesis, and sequencing by ligation.
 4. The method according to claim 1, wherein the methylation of the SDC2 gene is detected by detecting a material that binds to the probe and exhibits fluorescence.
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